Sang Yoon Lee

MSc Robotics, School of Engineering Mathematics and Technology, University of Bristol
BEng Electronic Engineering, College of Engineering, Kyunghee University
Email: sangyoon4@gmail.com | Phone: +82-10-6555-2342, +44-777-461-3236

PERSONAL DATA

- ✓ Birth: 19th February, 1997, in Republic of Korea
- ✓ Nationality: Korean
- ✓ Language: Korean (native), English (fluent)
- ✓ Military Service: Honorably discharged as Sergeant (December 2018)

EDUCATION

Seb. 2025

University of Bristol (MSc Robotics) Bristol, United Kingdom

- Relevant Coursework: Machine Vision, Robotics Fundamentals, Advanced Control & Dynamics, Robot Learning (DMP/GMM), HRI, Assistive Robotics
- Master's Thesis: Spatialized Audio Neurofeedback-based Calibration of a Motor Imagery EEG-BCI for Robot Control (in progress)
- Current Average : 74% (Distinction)

Mar. 2016 ~ Feb. 2022

Kyunghee University Seoul, Korea

1st Bachelor of Electronic Engineering

GPA 3.39/4.5

RESEARCH EXPERIENCES

Master's Thesis Research

Spatialized Audio Neurofeedback-based Calibration of a Motor Imagery EEG-BCI for Robot Control

- Investigating alternative neurofeedback modalities to traditional visual feedback for motor imagery BCI calibration, reducing cognitive load and attentional resource competition
- Designing and implementing a 5-phase experimental protocol with 30 BCI-naive participants comparing discrete vs. continuous spatialized audio feedback paradigms against visual control
- Developing 16-channel EEG signal processing pipeline with focus on C3/C4 sensorimotor sites for left/right hand motor imagery classification
- Validating neurofeedback effectiveness through progressive testing: volitional control assessment, simulated robotic arm pick-and-place tasks, and physical 1-DOF robotic arm control
- Hypothesizing that spatialized audio feedback will match or exceed visual feedback accuracy with lower cognitive workload, while discrete feedback will yield more robust transfer to robotic control applications

Mobile Robot Navigation & Control

- Developed a decentralized leader-follower system on differential-drive robots (Pololu 3pi+) using C++, implementing IR-based relative localization for formation control without external infrastructure.
- Engineered a dynamic role-switching FSM (Finite State Machine) triggered by IMU-based collision detection, enabling the formation to autonomously reverse leadership and escape dead-ends where traditional backtracking failed.

- Implemented PID control with signal processing filters (SMA, EMA) to mitigate sensor noise, achieving a >95% success rate in narrow corridors and reducing total travel distance by 27.1%.

Advanced Control & Dynamics

- Modelled and controlled a two-link planar manipulator in Simulink, implementing computed torque control with state-space linearisation and Lyapunov stability analysis
- Designed PD and PID controllers with gravity compensation for joint-space trajectory tracking, evaluating robustness under parameter uncertainty and external disturbances

Robot Learning: DMP & GMM-based Motion Generation

- Implemented Dynamic Movement Primitives (DMP) for learning and reproducing demonstrated trajectories with tuneable goal adaptation and temporal scaling in MATLAB
- Applied Gaussian Mixture Models (GMM) with Gaussian Mixture Regression (GMR) for probabilistic trajectory encoding and generalisation across varying task conditions

ML-based Autonomous Robot Navigation

- Developed a Python-based ML pipeline for autonomous robot maze navigation, integrating sensor fusion with learned decision policies for obstacle avoidance and path planning
- Trained and evaluated reinforcement learning and supervised learning models for real-time motion planning under dynamic environmental constraints

WORK EXPERIENCES

Worked as a PMIC Application Engineer at LX Semicon (May. 2022~ Aug. 2025)

- Developed Python-based automation programs for measurement instruments (oscilloscopes, DMMs, power supplies) using GPIB communication, reducing efficiency measurement evaluation time
- Conducted comprehensive PMIC characteristic analysis including BUCK/BOOST efficiency measurements, Line/Load regulation evaluation, and IP current consumption analysis
- Designed and produced custom PCB boards for PMIC testing and competitor evaluation using OrCad for circuit design and PCB artwork/routing
- Performed system-level analysis of PMIC power sequences and LS CLK characteristics on SOC, including defect analysis and RMA response using automated data processing

RESEARCH INTEREST

Brain-Computer Interfaces & Neurotechnology

- Motor imagery EEG signal processing, neurofeedback paradigms, BCI-based robotic control

Control Theory & Applications

- Optimal control, robust control, intelligent control, multi-sensor fusion

Robotics & Mechatronics

- Humanoid robots, mobile robots, multi-robot coordination, human-robot interaction

Computer Vision & Machine Learning

- Image processing using machine learning, visual perception for robotics

TECHNICAL STRENGTHS

Programming Languages

- C/C++, C#, Python, MATLAB/Simulink, Verilog (HDL)

Robotics & Control

- Machine Vision, EEG signal processing, BCI systems, robotic arm control, DMP/GMM, ROS2

Hardware & Electronics

- PCB design and artwork (OrCad), soldering, MCU interface systems (Atmega 128, MSP430 , STM32 Nucleo)

Measurement & Instrumentation

- GPIB automation, oscilloscopes, DMMs, power supply control, I2C protocol

AWARDS AND HONORS

- CREATIVE COMPREHENSIVE DESIGN COMPETITION AWARD, KYUNGHEE UNIVERSITY (2021)

LEADERSHIP & ACTIVITIES

- President, Micro Robot Academic Club, Kyunghee University (2021)
- Vice President, College of Electronics and Information, Kyunghee University (2022)
- Coding Instructor, Ondream School Davinci Class for elementary students (2022)
- D-Buddy Supporter, Daejeon Foreign Resident Center (2024)

REFEREES

Professor Giacinto Barresi (MSc Dissertation Supervisor)

- Professor, College of Arts, Technology and Environment School of Engineering
- Bristol Robotics Laboratory, University of Bristol
- Email: Giacinto.Barresi@uwe.ac.uk
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Professor Dong-han Kim (Bachelor Dissertation Supervisor)

- Professor, Department of Electronics Engineering
- College of Electronics and Information, Kyung Hee University
- Email: dhkim@khu.ac.kr